Table of Contents

ENVIRONMENT SETUP 1
BASE ENVIRONMENT CREATION 2
NEW DIRECTORY 2
INSTALL PYTHON 3
CREATE A VIRTUAL ENVIRONMENT 7
INSTALL PACKAGES 7
INSTALL NODEJS 10
CREATE ENV SCRIPT 12
INSTALL OLLAMA PACKAGES WITH NPM 14
INSTALL OLLAMA 15
CREATE APPLICATION LAUNCHER SCRIPT 26
STEPS TO RUN THE SERVER/CLIENT 27

ENVIRONMENT SETUP

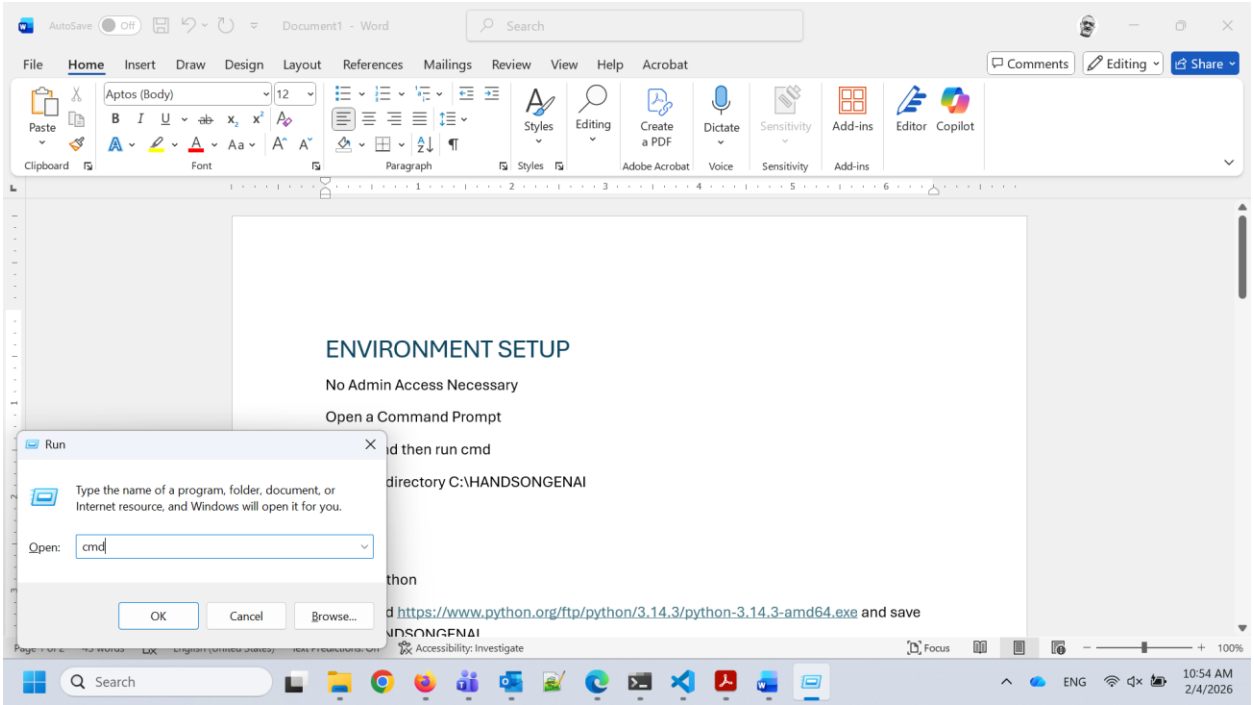
To do the hands on exercises and get one familiar with invoking an LLM in a programming environment, this environment setup – windows admin access is not necessary – section helps one setup base programming tools – Python & NodeJS

BASE ENVIRONMENT CREATION

NEW DIRECTORY

Open a Command Prompt

Win+R and then run cmd



Create a directory C:\HANDSONGENAI

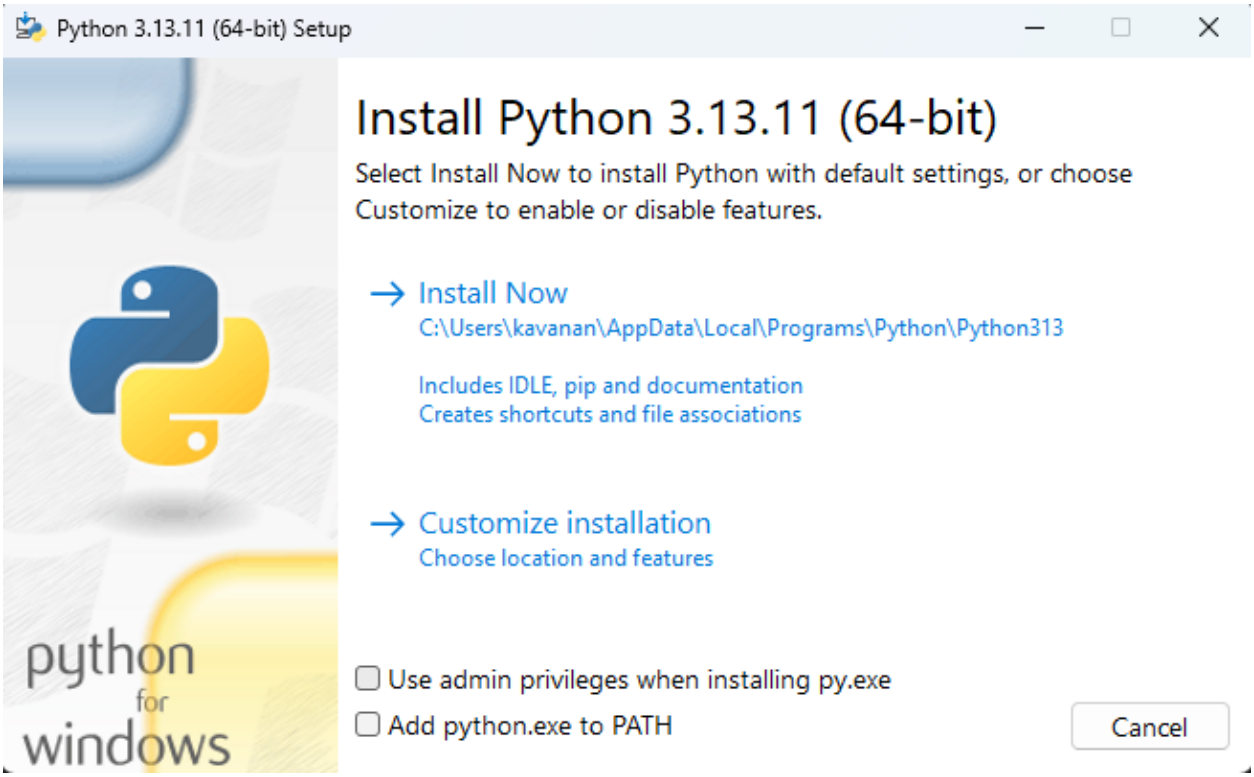
```
Command Prompt

Microsoft Windows [Version 10.0.22631.2861]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kavanan>cd C:\

C:\>mkdir HANDSONGENAI

C:\>
```

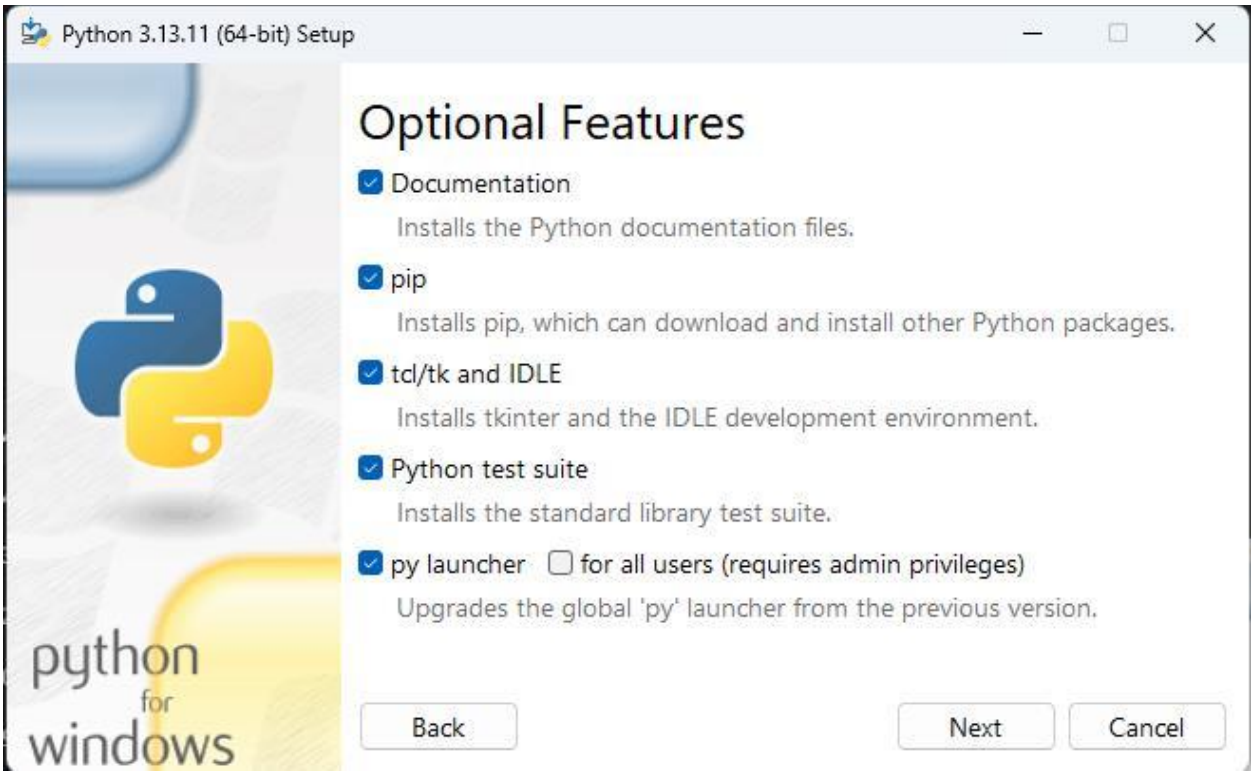
INSTALL PYTHON

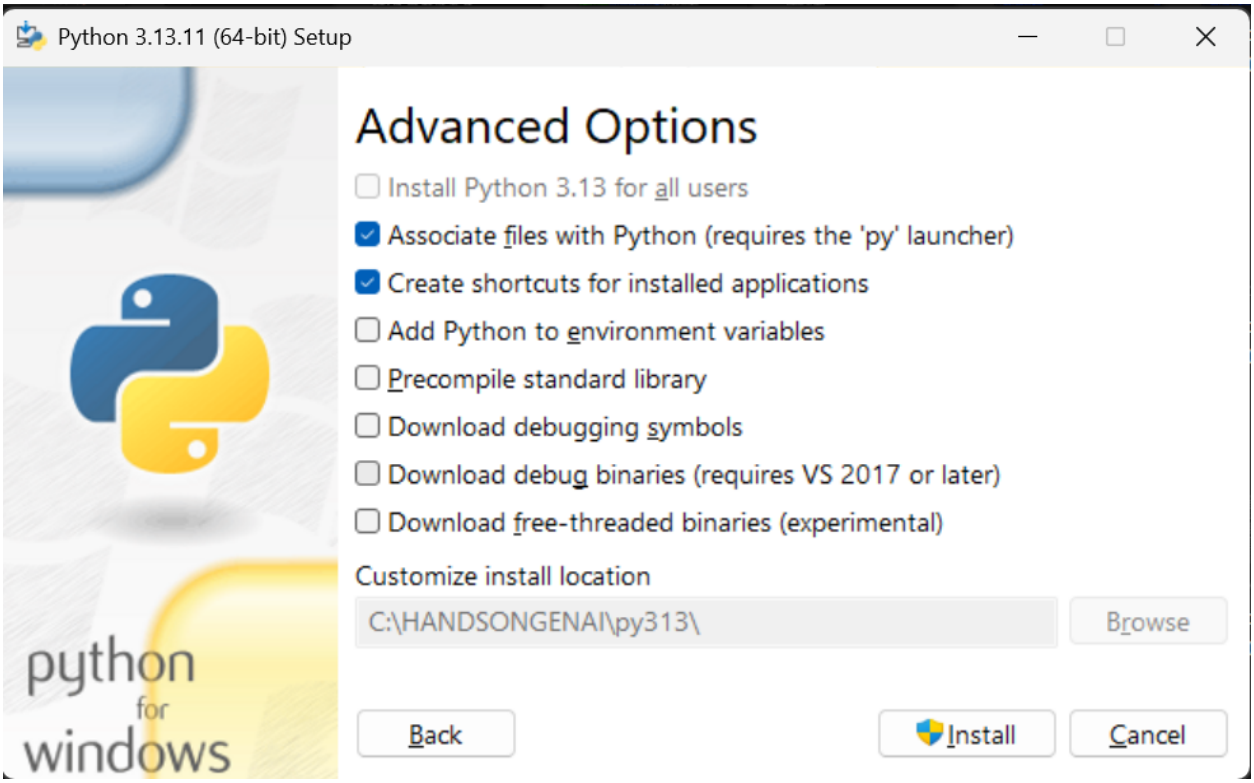
Download https://www.python.org/ftp/python/3.13.11/python-3.13.11-amd64.exe and save to C:\HANDSONGENAI

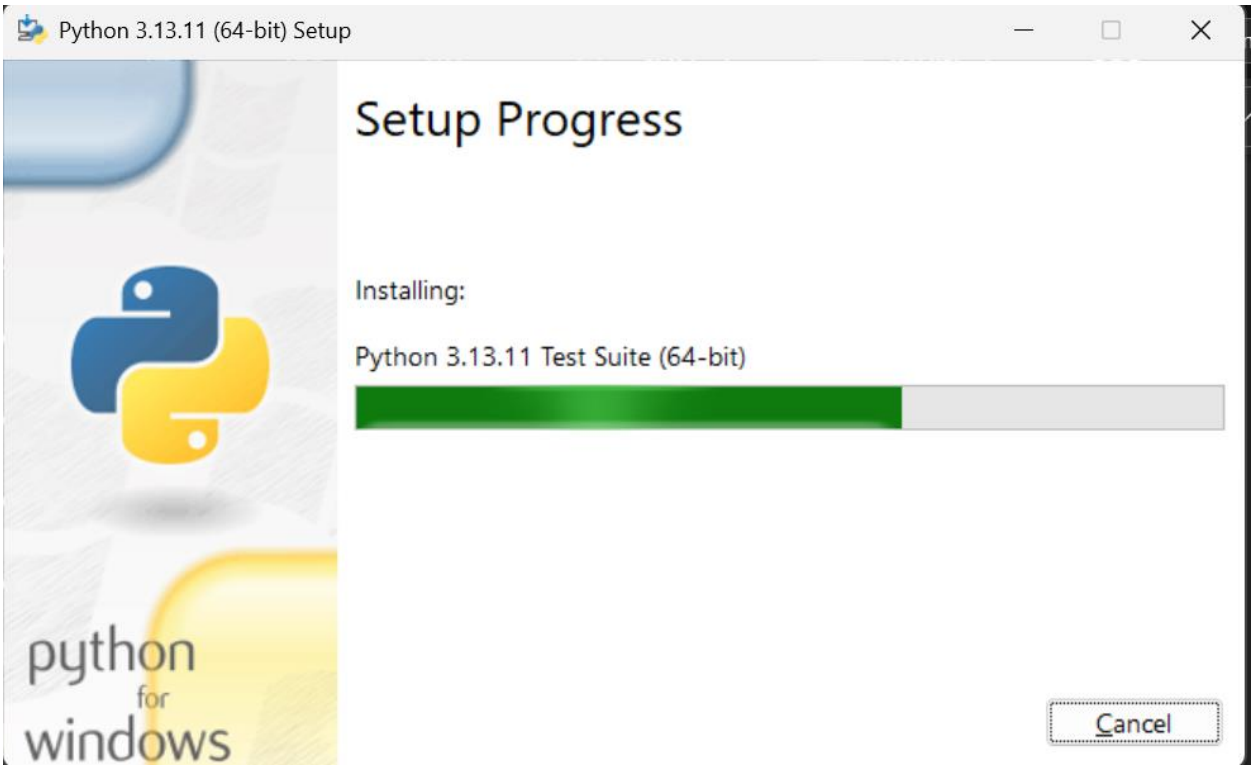
```
Command Prompt

Microsoft Windows [Version 10.0.22631.2861]
(c) Microsoft Corporation. All rights reserved.

C:\Users\kavanan>cd HANDSONGENAI
The system cannot find the path specified.

C:\Users\kavanan>cd C:\

C:\>cd HANDSONGENAI

C:\HANDSONGENAI>dir
 Volume in drive C has no label.
 Volume Serial Number is 6064-283B

 Directory of C:\HANDSONGENAI

02/04/2026  03:13 PM    <DIR>          .
02/04/2026  02:40 PM        28,775,744 python-3.13.11-amd64.exe
               1 File(s)     28,775,744 bytes
               1 Dir(s)  450,670,313,472 bytes free

C:\HANDSONGENAI>
```

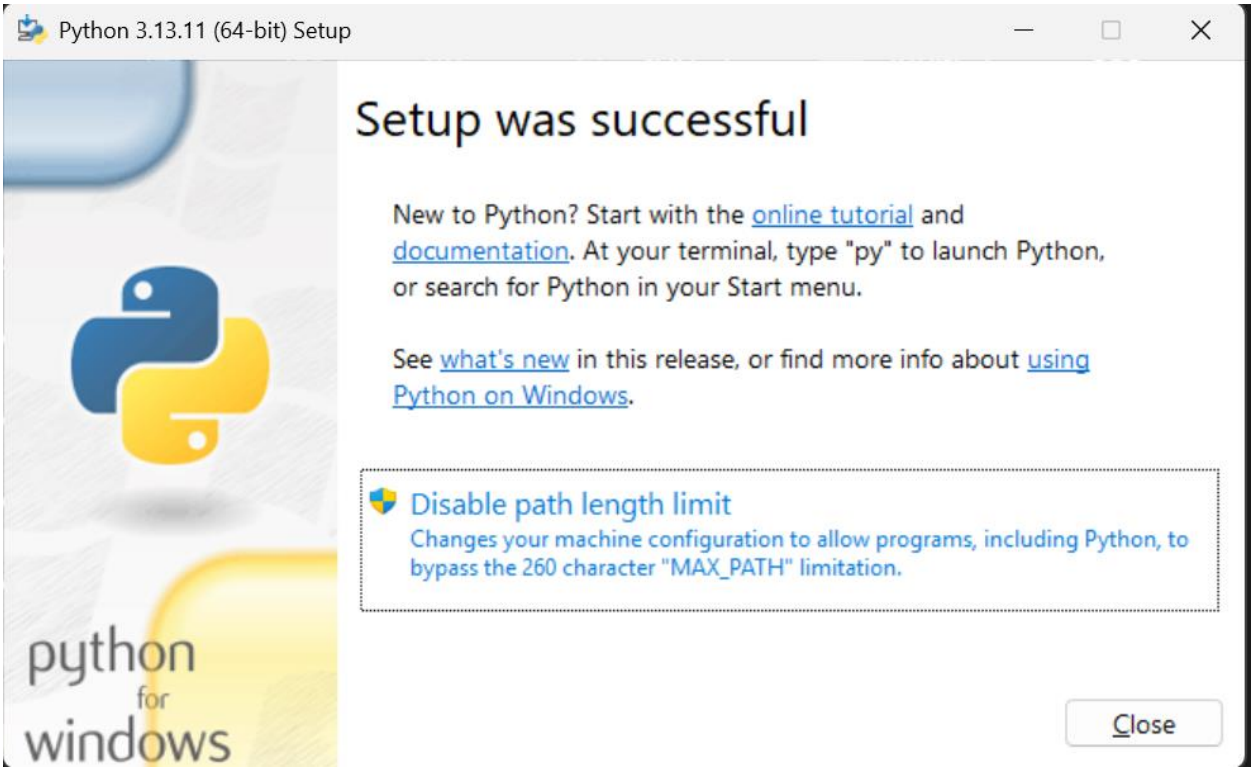
Run the downloaded executable to install Python to C:\HANDSONGENAI\py313. Unselect admin privileges and then in the next screen make the installation directory to be C:\HANDSONGENAI\py313




Python 3.13.11 (64-bit) Setup
Optional Features
Documentation
Installs the Python documentation files.
pip
Installs pip, which can download and install other Python packages.
tcl/tk and IDLE
Installs tkinter and the IDLE development environment.
Python test suite
Installs the standard library test suite.
py launcher
for all users (requires admin privileges)
Upgrades the global 'py' launcher from the previous version.
python
for
windows
Back
Next
Cancel


Python 3.13.11 (64-bit) Setup
Advanced Options
Install Python 3.13 for all users
Associate files with Python (requires the 'py' launcher)
Create shortcuts for installed applications
Add Python to environment variables
Precompile standard library
Download debugging symbols
Download debug binaries (requires VS 2017 or later)
Download free-threaded binaries (experimental)
Customize install location
C:\HANDSONGENAI\py313\
Browse
python
for
windows
Back
Install
Cancel





Verify python 3.13 is installed as below

```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\HANDSONGENAI>dir
 Volume in drive C has no label.
 Volume Serial Number is E686-4FAE

 Directory of C:\HANDSONGENAI

04-02-2026  15:23    <DIR>          .
04-02-2026  15:07    <DIR>          py313
04-02-2026  14:43        28,775,744 python-3.13.11-amd64.exe
               1 File(s)     28,775,744 bytes
               2 Dir(s)  327,779,180,544 bytes free

C:\HANDSONGENAI>py313\python.exe -V
Python 3.13.11

C:\HANDSONGENAI>py313\python.exe
Python 3.13.11 (tags/v3.13.11:6278944, Dec  5 2025, 16:26:58) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("Hello!")
Hello!
>>> import os
>>> import sys
>>> print(sys.path)
['', 'C:\\HANDSONGENAI\\py313\\python313.zip', 'C:\\HANDSONGENAI\\py313\\DLLs', 'C:\\HANDSONGENAI\\py313\\Lib', 'C:\\HANDSONGENAI\\py
313', 'C:\\HANDSONGENAI\\py313\\Lib\\site-packages']
>>> exit

C:\HANDSONGENAI>
```

CREATE A VIRTUAL ENVIRONMENT

Execute the commands as below (`py313\python -m venv vpy313`) to create the virtual environment and activate (`vpy313\scripts\activate`) the virtual environment. Verify python in the virtual environment works

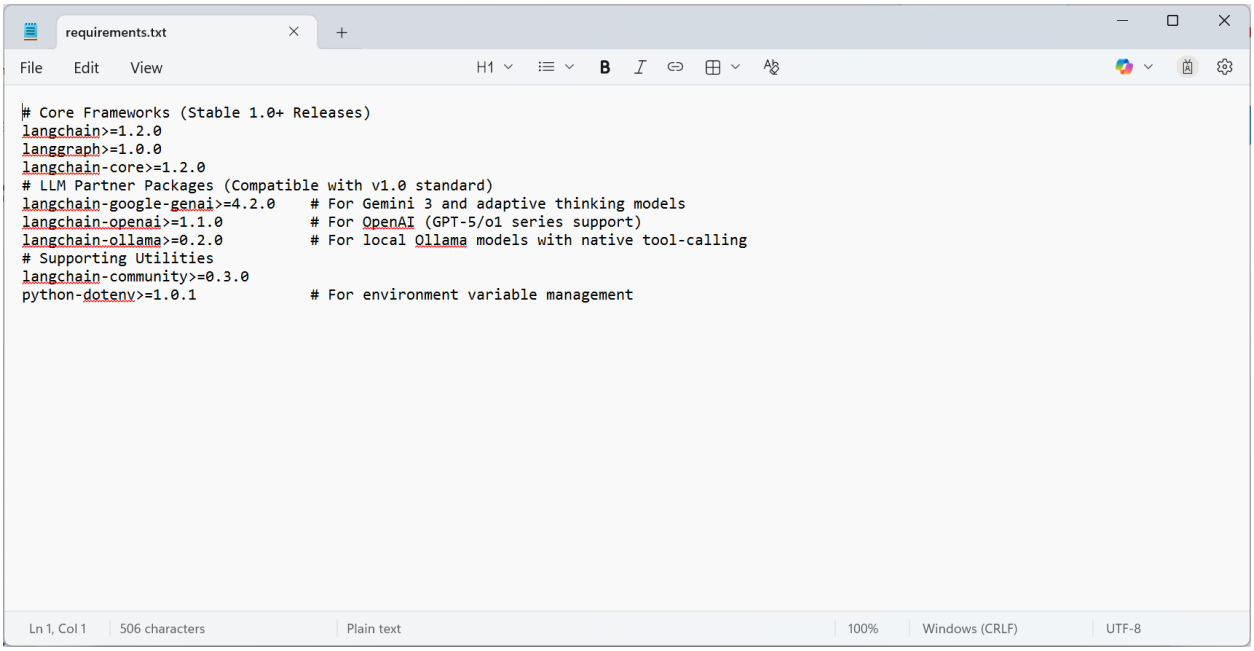
```
C:\Windows\System32\cmd.e

>>> exit

C:\HANDSONGENAI>py313\python.exe -m venv vpy313

C:\HANDSONGENAI>dir
 Volume in drive C has no label.
 Volume Serial Number is E686-4FAE

 Directory of C:\HANDSONGENAI

04-02-2026  15:29    <DIR>          .
04-02-2026  15:07    <DIR>          py313
04-02-2026  14:43        28,775,744 python-3.13.11-amd64.exe
04-02-2026  15:29    <DIR>          vpy313
               1 File(s)     28,775,744 bytes
               3 Dir(s)  327,786,045,440 bytes free

C:\HANDSONGENAI>vpy313\Scripts\activate

(vpy313) C:\HANDSONGENAI>python -V
Python 3.13.11

(vpy313) C:\HANDSONGENAI>python
Python 3.13.11 (tags/v3.13.11:6278944, Dec  5 2025, 16:26:58) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import sys
>>> print(sys.path)
['', 'C:\\HANDSONGENAI\\py313\\python313.zip', 'C:\\HANDSONGENAI\\py313\\DLLs', 'C:\\HANDSONGENAI\\py313\\Lib', 'C:\\HANDSONGENAI\\py
313', 'C:\\HANDSONGENAI\\vpy313', 'C:\\HANDSONGENAI\\vpy313\\Lib\\site-packages']
>>>
```

INSTALL PACKAGES

Create a `requirements.txt` file in C:\HANDSONGENAI directory. The contents are as below

```
# Core Frameworks (Stable 1.0+ Releases)
langchain>=1.2.0
langgraph>=1.0.0
```

```
langchain-core>=1.2.0
# LLM Partner Packages (Compatible with v1.0 standard)
langchain-google-genai>=4.2.0    # For Gemini 3 and adaptive thinking models
langchain-openai>=1.1.0          # For OpenAI (GPT-5/o1 series support)
langchain-ollama>=0.2.0          # For local Ollama models with native tool-calling
# Supporting Utilities
langchain-community>=0.3.0
python-dotenv>=1.0.1             # For environment variable management
```



Run this pip command to install the packages

```
pip install -r requirements.txt --disable-pip-version-check
```

```
C:\Windows\System32\cmd.ex   Command Prompt

(vpy313) C:\HANDSONGENAI>pip install -r requirements.txt --disable-pip-version-check
Collecting langchain>=1.2.0 (from -r requirements.txt (line 3))
  Downloading langchain-1.2.8-py3-none-any.whl.metadata (5.0 kB)
Collecting langgraph>=1.0.0 (from -r requirements.txt (line 5))
  Downloading langgraph-1.0.7-py3-none-any.whl.metadata (7.4 kB)
Collecting langchain-core>=1.2.0 (from -r requirements.txt (line 7))
  Downloading langchain_core-1.2.8-py3-none-any.whl.metadata (3.7 kB)
Collecting langchain-google-genai>=4.2.0 (from -r requirements.txt (line 11))
  Downloading langchain_google_genai-4.2.0-py3-none-any.whl.metadata (2.7 kB)
Collecting langchain-openai>=1.1.0 (from -r requirements.txt (line 13))
  Downloading langchain_openai-1.1.7-py3-none-any.whl.metadata (2.6 kB)
Collecting langchain-ollama>=0.2.0 (from -r requirements.txt (line 15))
  Downloading langchain_ollama-1.0.1-py3-none-any.whl.metadata (2.5 kB)
Collecting langchain-community>=0.3.0 (from -r requirements.txt (line 19))
  Downloading langchain_community-0.4.1-py3-none-any.whl.metadata (3.0 kB)
Collecting python-dotenv>=1.0.1 (from -r requirements.txt (line 21))
  Downloading python_dotenv-1.2.1-py3-none-any.whl.metadata (25 kB)
Collecting pydantic<3.0.0,>=2.7.4 (from langchain>=1.2.0->-r requirements.txt (line 3))
  Downloading pydantic-2.12.5-py3-none-any.whl.metadata (90 kB)
Collecting langgraph-checkpoint<5.0.0,>=2.1.0 (from langgraph>=1.0.0->-r requirements.txt (line 5))
  Downloading langgraph_checkpoint-4.0.0-py3-none-any.whl.metadata (4.9 kB)
Collecting langgraph-prebuilt<1.1.0,>=1.0.7 (from langgraph>=1.0.0->-r requirements.txt (line 5))
  Downloading langgraph_prebuilt-1.0.7-py3-none-any.whl.metadata (5.2 kB)
Collecting langgraph-sdk<0.4.0,>=0.3.0 (from langgraph>=1.0.0->-r requirements.txt (line 5))
  Downloading langgraph_sdk-0.3.3-py3-none-any.whl.metadata (1.6 kB)
Collecting xxhash>=3.5.0 (from langgraph>=1.0.0->-r requirements.txt (line 5))
  Downloading xxhash-3.6.0-cp313-cp313-win_amd64.whl.metadata (13 kB)
Collecting jsonpatch<2.0.0,>=1.33.0 (from langchain-core>=1.2.0->-r requirements.txt (line 7))
  Using cached jsonpatch-1.33-py2.py3-none-any.whl.metadata (3.0 kB)
Collecting langsmith<1.0.0,>=0.3.45 (from langchain-core>=1.2.0->-r requirements.txt (line 7))
  Downloading langsmith-0.6.8-py3-none-any.whl.metadata (15 kB)
Collecting packaging>=23.2.0 (from langchain-core>=1.2.0->-r requirements.txt (line 7))
```

```
C:\Windows\System32\cmd.ex   Command Prompt

Downloading propcache-0.4.1-cp313-cp313-win_amd64.whl (40 kB)
Downloading pyasn1-0.6.2-py3-none-any.whl (83 kB)
Using cached pyasn1_modules-0.4.2-py3-none-any.whl (181 kB)
Downloading regex-2026.1.15-cp313-cp313-win_amd64.whl (277 kB)
Using cached requests_toolbelt-1.0.0-py2.py3-none-any.whl (54 kB)
Downloading tqdm-4.67.3-py3-none-any.whl (78 kB)
Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
Downloading xxhash-3.6.0-cp313-cp313-win_amd64.whl (31 kB)
Using cached zstandard-0.25.0-cp313-cp313-win_amd64.whl (506 kB)
Using cached colorama-0.4.6-py2.py3-none-any.whl (25 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Using cached sniffio-1.3.1-py3-none-any.whl (10 kB)
Installing collected packages: filetype, zstandard, xxhash, websockets, uuid-utils, urllib3, typing-extensions, tenacity, sniffio, re
gex, pyyaml, python-dotenv, pycparser, pyasn1, propcache, packaging, ormsgpack, orjson, numpy, mypy-extensions, multidict, jsonpointe
r, jiter, idna, httpx-sse, h11, greenlet, frozenlist, distro, colorama, charset_normalizer, certifi, attrs, annotated-types, aiohappy
eyeballs, yarl, typing-inspection, typing-inspect, tqdm, SQLAlchemy, rsa, requests, pydantic-core, pyasn1-modules, marshmallow, jsonp
atch, httpcore, cffi, anyio, aiosignal, tiktoken, requests-toolbelt, pydantic, httpx, dataclasses-json, cryptography, aiohttp, pydant
ic-settings, openai, ollama, langsmith, langgraph-sdk, google-auth, langchain-core, langgraph-checkpoint, langchain-text-splitters, l
angchain-openai, langchain-ollama, google-genai, langgraph-prebuilt, langchain-google-genai, langchain-classic, langgraph, langchain-
community, langchain
Successfully installed SQLAlchemy-2.0.46 aiohappyeyeballs-2.6.1 aiohttp-3.13.3 aiosignal-1.4.0 annotated-types-0.7.0 anyio-4.12.1 att
rs-25.4.0 certifi-2026.1.4 cffi-2.0.0 charset_normalizer-3.4.4 colorama-0.4.6 cryptography-46.0.4 dataclasses-json-0.6.7 distro-1.9.0
 filetype-1.2.0 frozenlist-1.8.0 google-auth-2.48.0 google-genai-1.62.0 greenlet-3.3.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 httpx-s
se-0.4.3 idna-3.11 jiter-0.13.0 jsonpatch-1.33 jsonpointer-3.0.0 langchain-1.2.8 langchain-classic-1.0.1 langchain-community-0.4.1 la
ngchain-core-1.2.8 langchain-google-genai-4.2.0 langchain-ollama-1.0.1 langchain-openai-1.1.7 langchain-text-splitters-1.1.0 langgrap
h-1.0.7 langgraph-checkpoint-4.0.0 langgraph-prebuilt-1.0.7 langgraph-sdk-0.3.3 langsmith-0.6.8 marshmallow-3.26.2 multidict-6.7.1 my
py-extensions-1.1.0 numpy-2.4.2 ollama-0.6.1 openai-2.16.0 orjson-3.11.7 ormsgpack-1.12.2 packaging-26.0 propcache-0.4.1 pyasn1-0.6.2
 pyasn1-modules-0.4.2 pycparser-3.0 pydantic-2.12.5 pydantic-core-2.41.5 pydantic-settings-2.12.0 python-dotenv-1.2.1 pyyaml-6.0.3 re
gex-2026.1.15 requests-2.32.5 requests-toolbelt-1.0.0 rsa-4.9.1 sniffio-1.3.1 tenacity-9.1.2 tiktoken-0.12.0 tqdm-4.67.3 typing-exten
sions-4.15.0 typing-inspect-0.9.0 typing-inspection-0.4.2 urllib3-2.6.3 uuid-utils-0.14.0 websockets-15.0.1 xxhash-3.6.0 yarl-1.22.0
zstandard-0.25.0

(vpy313) C:\HANDSONGENAI>
```

pysanity.py requirements.txt

File Edit View

```python
from langchain_google_genai import ChatGoogleGenerativeAI
from langchain_openai import ChatOpenAI
from langchain_ollama import ChatOllama
from langgraph.graph import StateGraph, MessagesState


#Select your provider
#model = ChatGoogleGenrativeAI(model="gemini-3-flash")
#model = ChatOpenAI(model="gpt-5")
model = ChatOllama(model="Ollama3.3")


# Basic LangGraph 1.0 Node
def call_model(state: MessagesState):
    response = model.invoke(state["messages"])
    return {"messages": [response]}

# Compile Graph
builder = StateGraph(MessagesState)
builder.add_node("agent", call_model)
builder.set_entry_point("agent")
graph = builder.compile()

print("Graph initialized with v1.0 libraries!")
```

Ln 13, Col 27 692 characters Plain text 100% Windows (CRLF) UTF-8

C:\Windows\System32\cmd.ex Command Prompt

```
(vpy313) C:\HANDSONGENAI>python pysanity.py
Graph initialized with v1.0 libraries!

(vpy313) C:\HANDSONGENAI>dir
 Volume in drive C has no label.
 Volume Serial Number is E686-4FAE

 Directory of C:\HANDSONGENAI

05-02-2026  11:49    <DIR>          .
04-02-2026  15:07    <DIR>          py313
05-02-2026  12:08               715 pysanity.py
04-02-2026  14:43        28,775,744 python-3.13.11-amd64.exe
05-02-2026  11:33               546 requirements.txt
04-02-2026  15:29    <DIR>          vpy313
               3 File(s)     28,777,005 bytes
               3 Dir(s)  326,262,779,904 bytes free

(vpy313) C:\HANDSONGENAI>
```

INSTALL NODEJS

Download https://nodejs.org/dist/v24.13.0/node-v24.13.0-win-x64.zip and save to C:\HANDSONGENAI. Unzip the contents of the downloaded zip to C:\HANDSONGENAI\nodejs. Now, nodejs software will be available as below

```
C:\Windows\System32\cmd.e

(vpy313) C:\HANDSONGENAI>cd nodejs

(vpy313) C:\HANDSONGENAI\nodejs>dir
 Volume in drive C is SSD OS
 Volume Serial Number is BAAE-9BA9

 Directory of C:\HANDSONGENAI\nodejs

09-02-2026  11:27    <DIR>          .
09-02-2026  11:37    <DIR>          ..
09-02-2026  11:27    <DIR>          node-v24.13.0-win-x64
               0 File(s)              0 bytes
               3 Dir(s)  24,203,862,016 bytes free

(vpy313) C:\HANDSONGENAI\nodejs>node-v24.13.0-win-x64
```

I suggest to rename the folder name to v24 to keep it short and verify npm works.

```
C:\Windows\System32\cmd.e

(vpy313) C:\HANDSONGENAI\nodejs>dir
 Volume in drive C is SSD OS
 Volume Serial Number is BAAE-9BA9

 Directory of C:\HANDSONGENAI\nodejs

09-02-2026  11:27    <DIR>          .
09-02-2026  11:37    <DIR>          ..
09-02-2026  11:27    <DIR>          node-v24.13.0-win-x64
               0 File(s)              0 bytes
               3 Dir(s)  24,288,796,672 bytes free

(vpy313) C:\HANDSONGENAI\nodejs>ren node-v24.13.0-win-x64 v24

(vpy313) C:\HANDSONGENAI\nodejs>v24\npm --help
```

```
C:\Windows\System32\cmd.e

npm install <foo>  add the <foo> dependency to your project
npm test           run this project's tests
npm run <foo>      run the script named <foo>
npm <command> -h   quick help on <command>
npm -l             display usage info for all commands
npm help <term>    search for help on <term> (in a browser)
npm help npm       more involved overview (in a browser)

All commands:

    access, adduser, audit, bugs, cache, ci, completion,
    config, dedupe, deprecate, diff, dist-tag, docs, doctor,
    edit, exec, explain, explore, find-dupes, fund, get, help,
    help-search, init, install, install-ci-test, install-test,
    link, ll, login, logout, ls, org, outdated, owner, pack,
    ping, pkg, prefix, profile, prune, publish, query, rebuild,
    repo, restart, root, run, sbom, search, set, shrinkwrap,
    star, stars, start, stop, team, test, token, undeprecate,
    uninstall, unpublish, unstar, update, version, view, whoami

Specify configs in the ini-formatted file:
    C:\Users\Admin\.npmrc
or on the command line via: npm <command> --key=value

More configuration info: npm help config
Configuration fields: npm help 7 config

npm@11.6.2 C:\HANDSONGENAI\nodejs\v24\node_modules\npm

(vpy313) C:\HANDSONGENAI\nodejs>
```

CREATE ENV SCRIPT

Create a *SETENV.cmd* in *C:\HANDSONGENAI* and paste the following lines of code in it:

```
@echo OFF
SET GENAIHOME=C:\HANDSONGENAI
SET NODEJS_HOME=%GENAIHOME%\nodejs\v24
SET PATH=%NODEJS_HOME%;%PATH%
if not exist vpy313 (
    echo Creating virtual environment...
    python -m venv vpy313
)
echo Activating Virtual Environment and calling Application to Launch.....
cmd /k "vpy313\Scripts\activate && APP.cmd"
pause
```

```
@echo OFF
SET GENAIHOME=C:\HANDSONGENAI
SET NODEJS_HOME=%GENAIHOME%\nodejs\v24
SET PATH=%NODEJS_HOME%;%PATH%
if not exist vpy313 (
    echo Creating virtual environment...
    python -m venv vpy313
)
echo Activating Virtual Environment and calling Application to Launch.....
cmd /k "vpy313\Scripts\activate && APP.cmd"

pause
```

Create a new folder “exercise” in the root path, and extract the nodejs24.zip file of the application within the “exercise” folder (assuming you have the nodejs24.zip file downloaded to your default downloads path:

```
mkdir exercise
tar -xf "%USERPROFILE%\Downloads\nodejs24.zip" -C "C:\HANDSONGENAI\exercise"
```

```
(vpy313) C:\HANDSONGENAI>mkdir exercise

(vpy313) C:\HANDSONGENAI>tar -xf "%USERPROFILE%\Downloads\nodejs24.zip" -C "C:\HANDSONGENAI\exercise"

(vpy313) C:\HANDSONGENAI>cd exercise

(vpy313) C:\HANDSONGENAI\exercise>dir
 Volume in drive C is SSD OS
 Volume Serial Number is BAAE-9BA9

 Directory of C:\HANDSONGENAI\exercise

09-02-2026  13:24    <DIR>          .
09-02-2026  13:24    <DIR>          ..
05-02-2026  16:16    <DIR>          nodejs24
               0 File(s)              0 bytes
               3 Dir(s)  24,467,980,288 bytes free

(vpy313) C:\HANDSONGENAI\exercise>
```

Quit the command prompt. Open a command prompt again and then change directory to C:\HANDSONGENAI and execute SETENV.cmd and verify that python and npm are available

```
npm config set strict-ssl false

(vpy313) C:\HANDSONGENAI>SETENV.cmd
Activating Virtual Environment and calling Application to Launch.....
-------Step 1-------
Creating folders...
A subdirectory or file chatapp already exists.
A subdirectory or file server already exists.
A subdirectory or file client already exists.
A subdirectory or file server\src already exists.
Getting Ready for Client Services
-------Step 2-------
Initializing Client in background...

Waiting for 0 seconds, press a key to continue ...
-------Step 3-------
Launching All Services via 03_run...
====================================
Starting backend server
====================================
====================================
Starting frontend (React)
====================================
====================================
All services launched
====================================
Press any key to continue . . .
Terminate batch job (Y/N)? y

(vpy313) C:\HANDSONGENAI\exercise\nodejs24\chatapp>cd ../../.. && npm --help
npm <command>

Usage:

npm install        install all the dependencies in your project
npm install <foo>  add the <foo> dependency to your project
npm test           run this project's tests
npm run <foo>      run the script named <foo>
npm <command> -h   quick help on <command>
```

INSTALL OLLAMA PACKAGES WITH NPM

Now, head to the application folder “nodejs24” where you have the chatapp files.

Open command prompt and change the directory to: "*C:\HANDSONGENAI\nodejs24\chatapp\server*" and type the following command.

```
npm install @langchain/ollama @langchain/core @langchain/langgraph --legacy-peer-deps
```

(or)

```
"C:\HANDSONGENAI\nodejs\v24\npm.cmd" install @langchain/ollama @langchain/core @langchain/langgraph --legacy-peer-deps
```

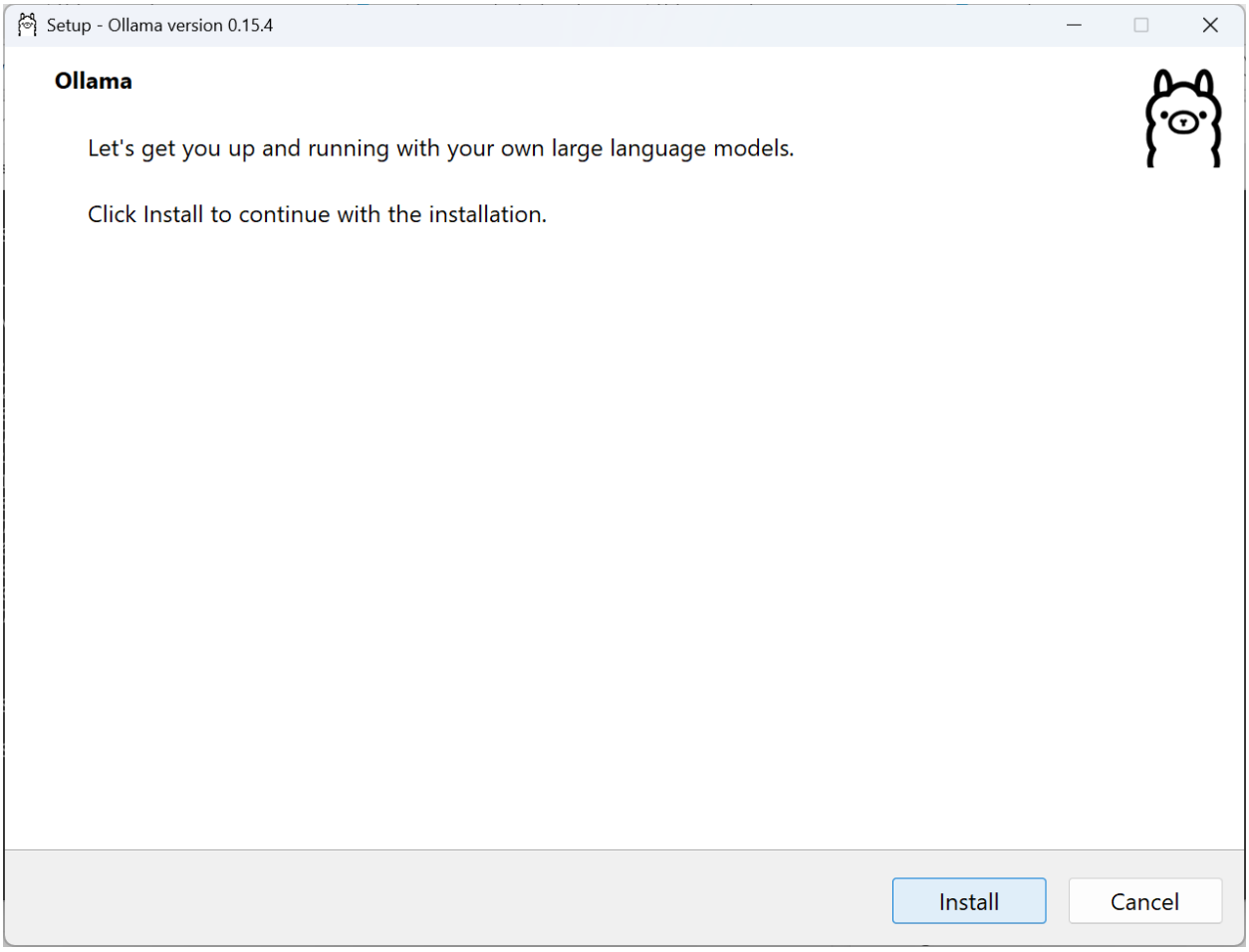
```
C:\Windows\System32\cmd.e

C:\HANDSONGENAI>cd C:\HANDSONGENAI\exercise\nodejs24\chatapp\server

C:\HANDSONGENAI\exercise\nodejs24\chatapp\server>"C:\HANDSONGENAI\nodejs\v24\npm.cmd" install @langchain/ollama @langcha
in/core @langchain/langgraph --legacy-peer-deps

up to date, audited 178 packages in 4s

34 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\HANDSONGENAI\exercise\nodejs24\chatapp\server>
```

INSTALL OLLAMA

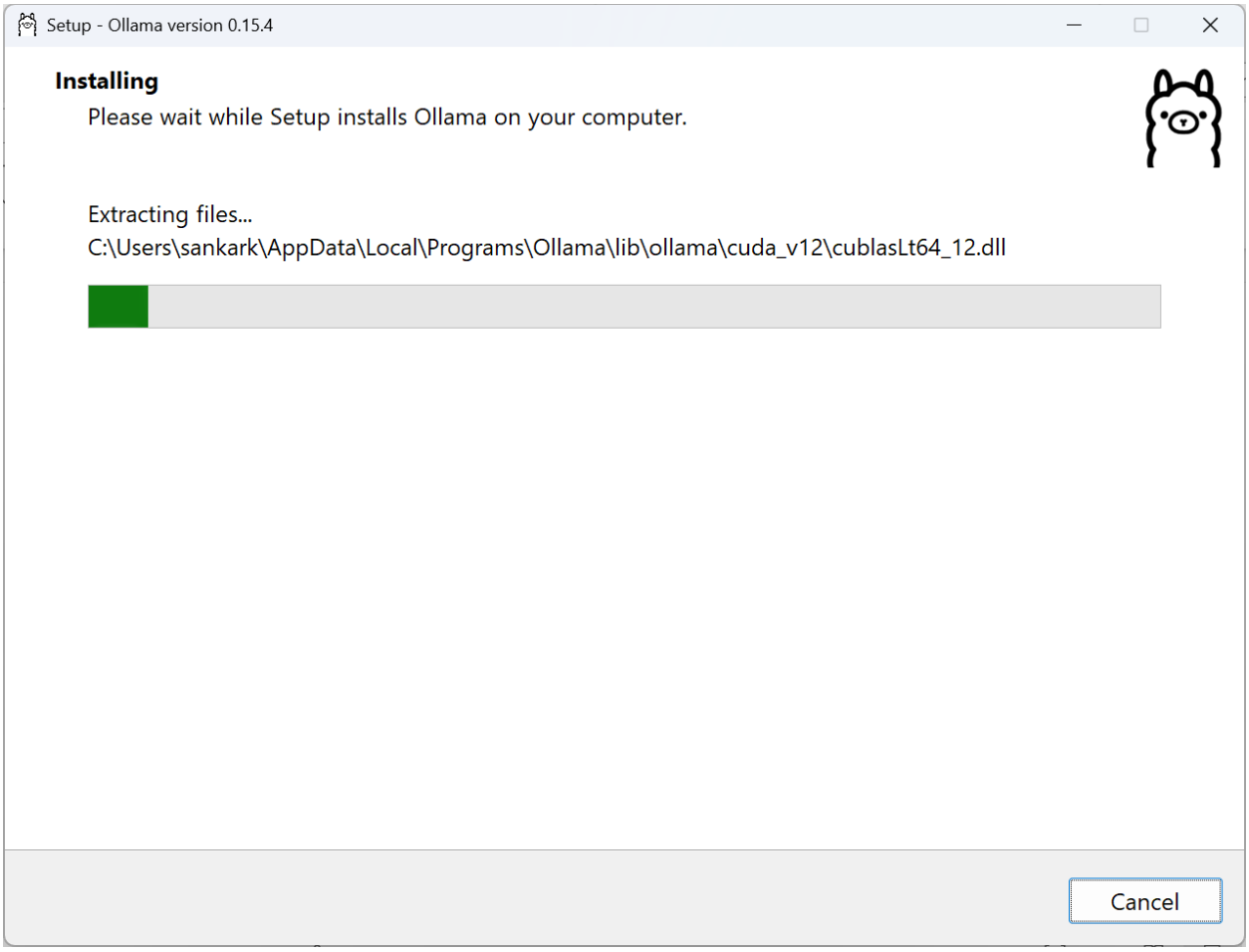
Download https://ollama.com/download/OllamaSetup.exe to C:\HANDSONGENAI

Then execute the downloaded file

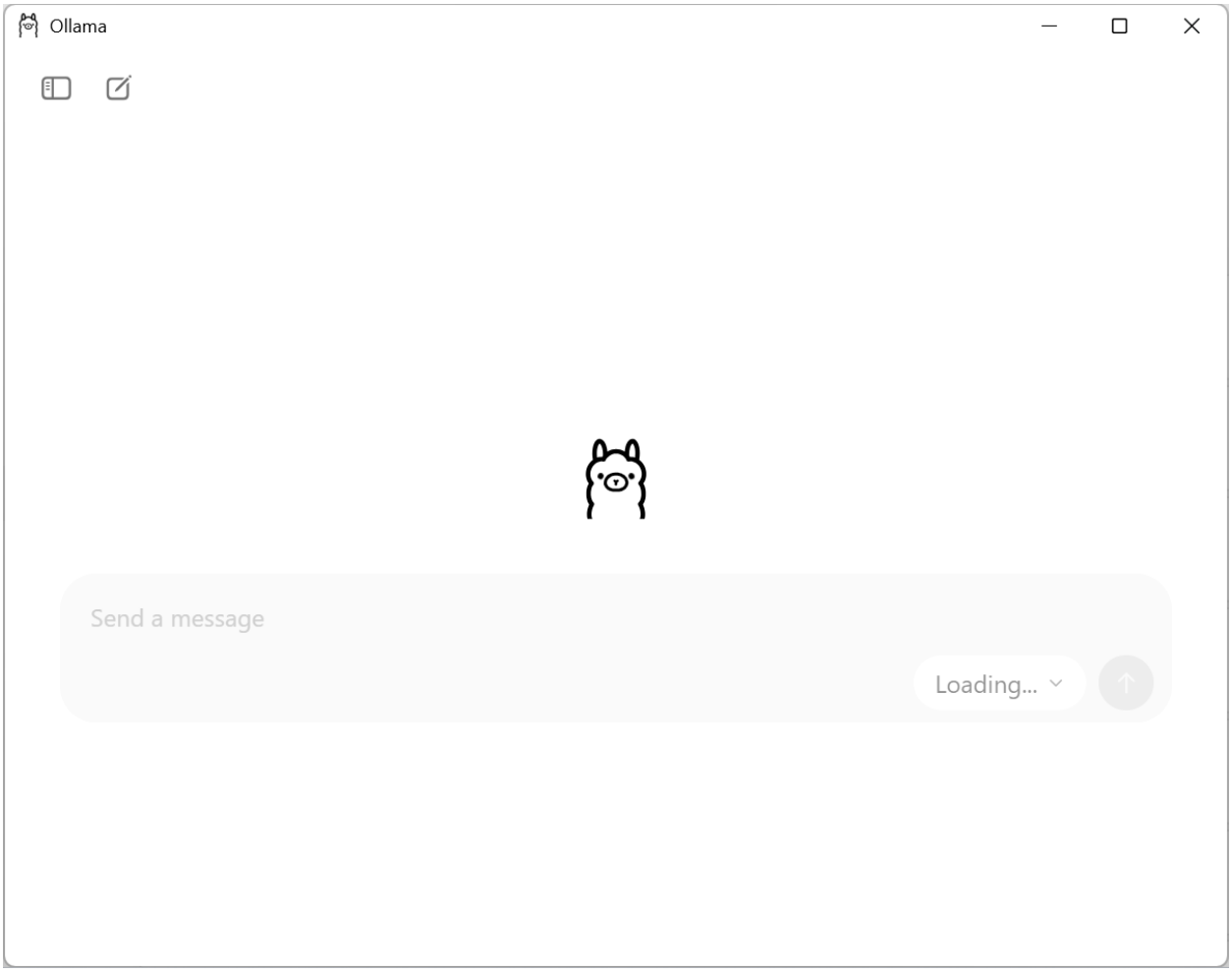
```
C:\Windows\System32\cmd.e

(vpy313) C:\HANDSONGENAI>dir
 Volume in drive C is SSD OS
 Volume Serial Number is BAAE-9BA9

 Directory of C:\HANDSONGENAI

09-02-2026  12:05    <DIR>          .
09-02-2026  11:23               669 APP.cmd
09-02-2026  11:40    <DIR>          nodejs
05-02-2026  16:16    <DIR>          nodejs24
09-02-2026  11:56             1,242 ollamachat.py
09-02-2026  11:54     1,262,567,096 OllamaSetup.exe
09-02-2026  11:23               688 pysanity.py
09-02-2026  11:23               516 requirements.txt
09-02-2026  11:34               325 SETENV.cmd
09-02-2026  12:05    <DIR>          vpy313
               6 File(s)  1,262,570,536 bytes
               4 Dir(s)  22,706,860,032 bytes free

(vpy313) C:\HANDSONGENAI>OllamaSetup.exe
```



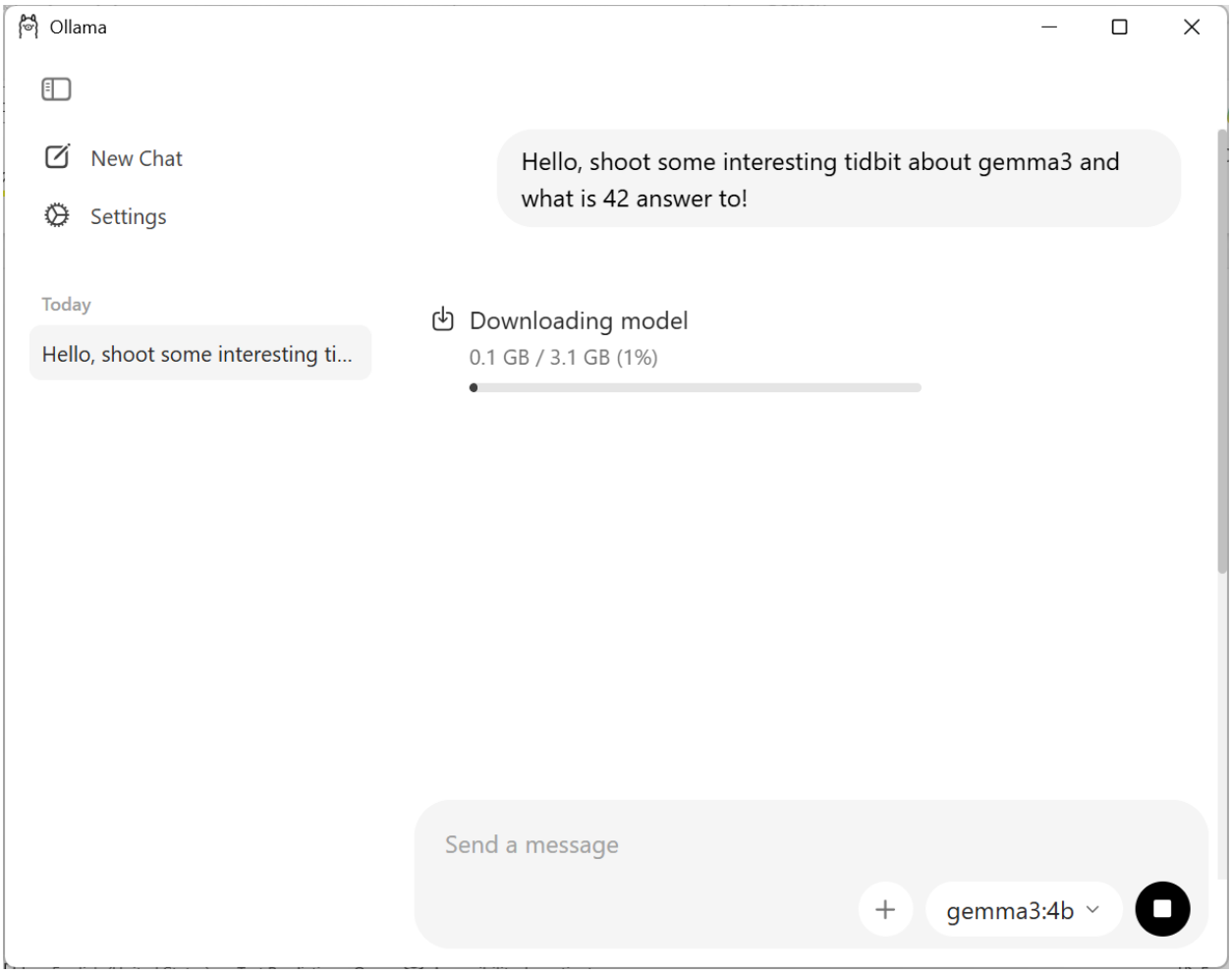
It will automatically install to the logged in users AppData\Local. Let it be so

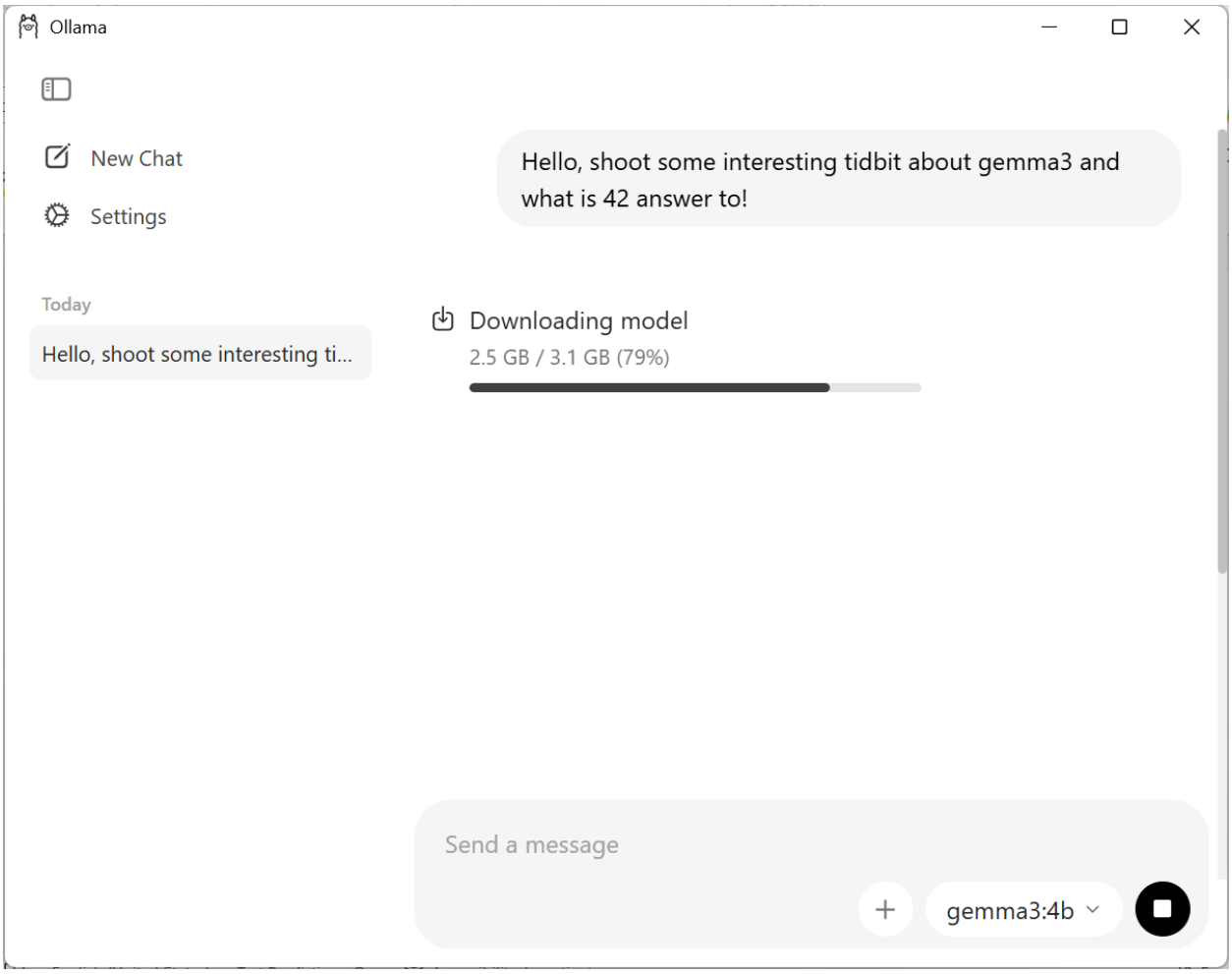


After installation, Ollama will automatically open up



If the UI seems unresponsive or stays “Loading..” quit and open after sometime. After that, select gemma3:4b as a model and ask it any question (example below). This should download the model – it will take some time depending on the network – and then result in an answer.


Ollama
New Chat
Settings
Today
Hello, shoot some interesting ti...
Hello, shoot some interesting tidbit about gemma3 and what is 42 answer to!
Downloading model
0.1 GB / 3.1 GB (1%)
Send a message
gemma3:4b



After downloading the model, it shall present the answer as below

Ollama

New Chat

Settings

Today

Hello, shoot some interesting ti...

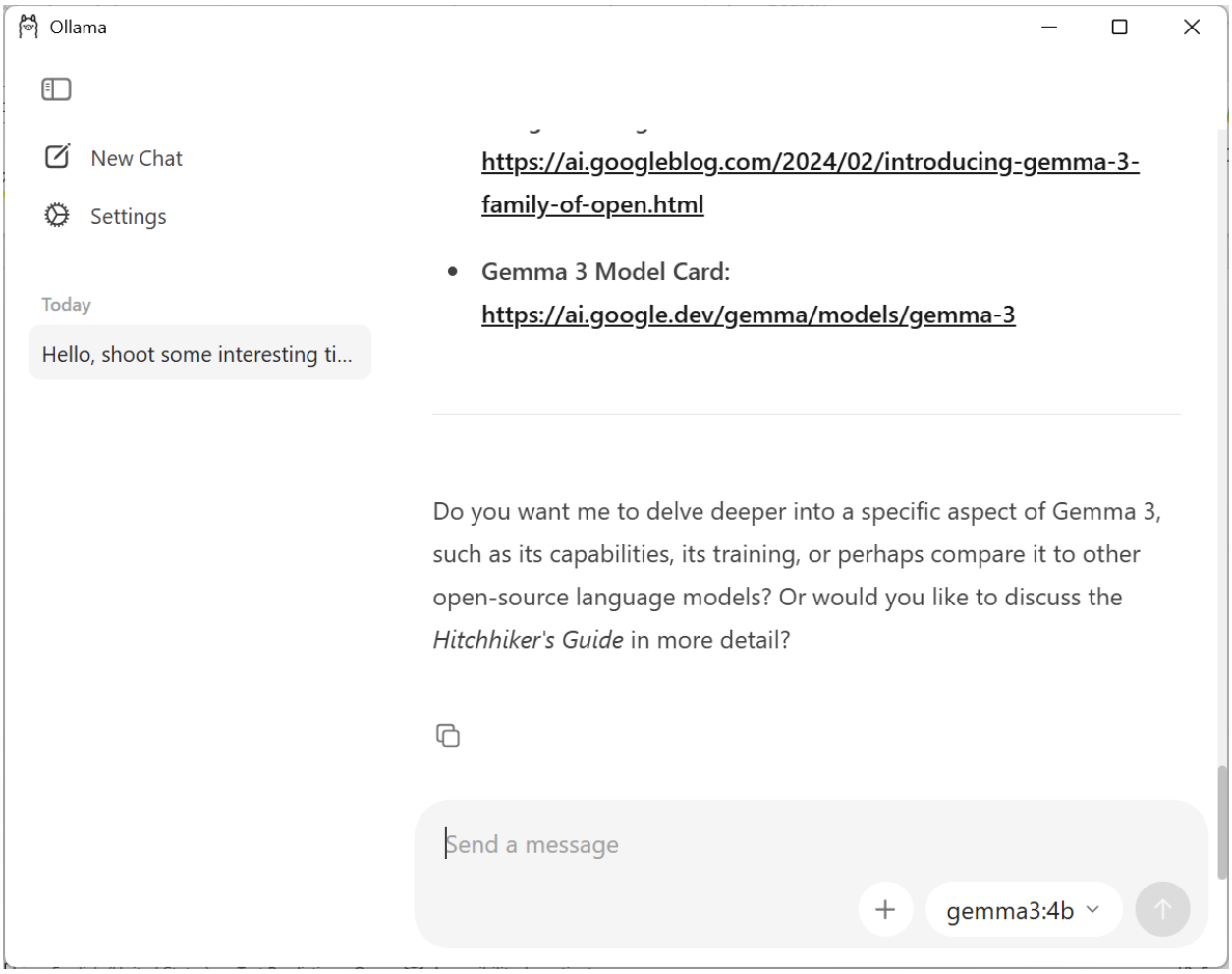
- **The Problem:** Of course, no one knows *what the question is*. Deep Thought explains it can only provide the answer, not the question.

- **The Further Explanation:** A second, even more powerful computer, Earth, is built to calculate the *question* that corresponds to 42. It runs for ten million years but ends abruptly before it can finish.

- **Douglas Adams' Intent:** Adams famously said he chose 42 completely at random. It's meant to be absurd and highlight the futility of

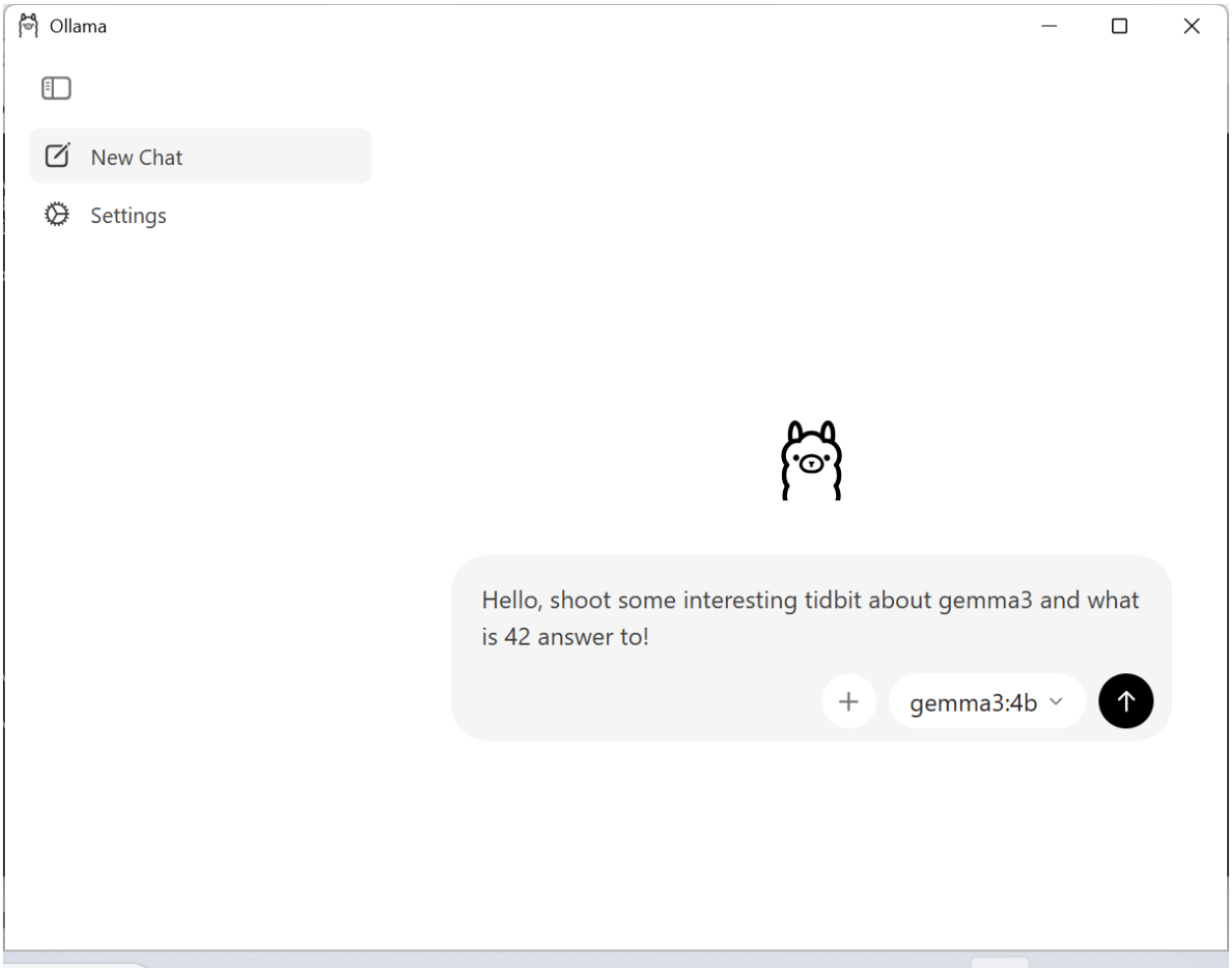
Send a message

gemma3:4b



You can also try with gemma3:1b which is a lower sized model. As the model sizes increase, one would need machines with GPU/VRAM or atleast 2-3x RAM of the model size. So if one downloads a 12b model, better to have atleast 24G+ RAM.

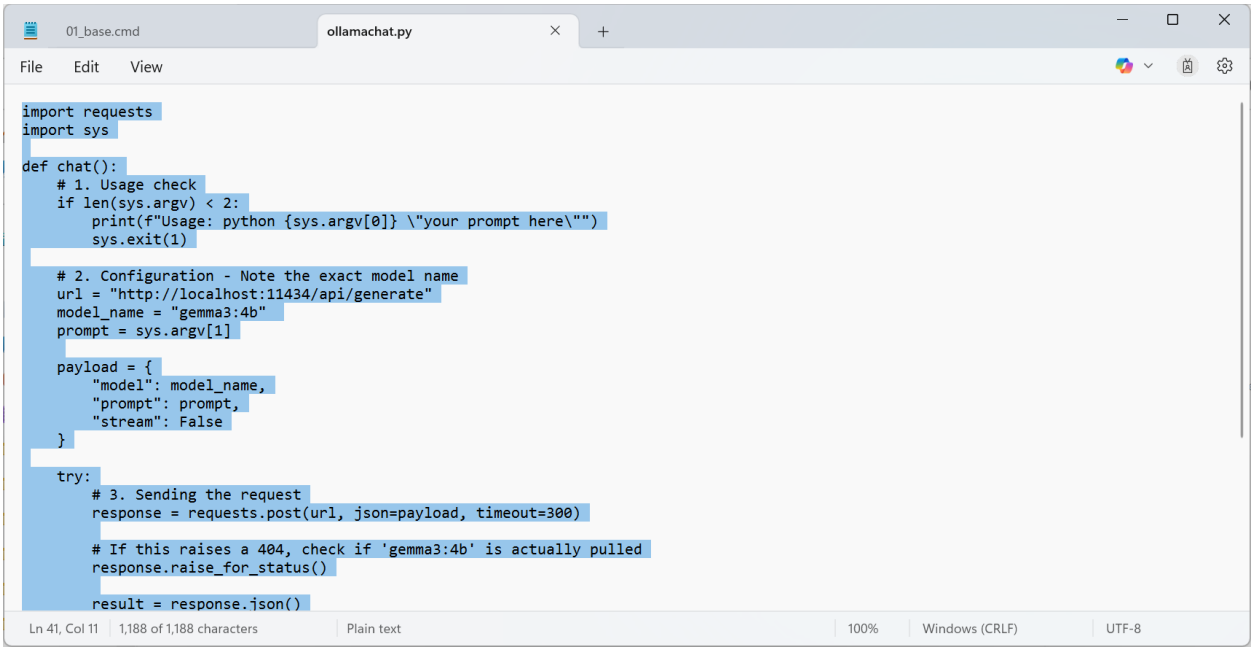
If all works and you see an answer that means the environment for playing around with programming in python or nodejs + local LLM (not very powerful) is ready.



To test ollama being accessible via HTTP api create a simple test program and run it. Create a new folder in exercise folder called “python” & paste the following code in the following location "C:\HANDSONGENAI\exercise\python\":

```
import requests
import sys

def chat():
    # 1. Usage check
    if len(sys.argv) < 2:
        print(f"Usage: python {sys.argv[0]} \"your prompt here\"")
        sys.exit(1)

    # 2. Configuration - Note the exact model name
    url = "http://localhost:11434/api/generate"
    model_name = "gemma3:4b"
```

```
    prompt = sys.argv[1]

    payload = {
        "model": model_name,
        "prompt": prompt,
        "stream": False
    }

    try:
        # 3. Sending the request
        response = requests.post(url, json=payload, timeout=300)

        # If this raises a 404, check if 'gemma3:4b' is actually pulled
        response.raise_for_status()

        result = response.json()
        print(result.get("response", "Error: No response text returned."))

    except requests.exceptions.HTTPError as e:
        if e.response.status_code == 404:
            print(f"Error 404: Endpoint not found or model '{model_name}' not pulled.")
            print("Try running: ollama pull gemma3:4b")
        else:
            print(f"HTTP Error: {e}")
    except Exception as e:
        print(f"An unexpected error occurred: {e}")

if __name__ == "__main__":
    chat()
```



Like below.

```
C:\Windows\System32\cmd.e

(vpy313) C:\HANDSONGENAI>cd exercise\python

(vpy313) C:\HANDSONGENAI\exercise\python>python ollamachat.py "What are the benefits of sleeping well?"
Okay, let's break down the incredible benefits of getting a good night's sleep! It's so much more than just feeling rest
ed - it's absolutely fundamental to your overall health and well-being. Here's a detailed look at the benefits, categori
zed for clarity:

**1. Physical Health Benefits:**

* **Boosts Immune System:** While you sleep, your body produces cytokines - proteins that help fight inflammation and in
fection. Chronic sleep deprivation severely weakens your immune response, making you more susceptible to colds, flu, and
 other illnesses.
* **Heart Health:**  Good sleep is linked to lower blood pressure, reduced risk of heart disease, and a healthier cardio
vascular system. Lack of sleep can contribute to inflammation, which plays a role in heart problems.
* **Weight Management:** Sleep regulates hormones that control appetite - specifically, ghrelin (which stimulates hunger
) and leptin (which signals fullness).  Insufficient sleep disrupts these hormones, leading to increased cravings, parti
cularly for unhealthy foods, and potentially weight gain.
* **Muscle Repair & Growth:** During deep sleep, your body releases growth hormone, which is crucial for muscle repair,
growth, and overall tissue maintenance.  This is particularly important for athletes and those engaging in regular exerc
ise.
* **Cell Repair & Regeneration:** Sleep allows your body to repair and regenerate cells throughout the body. This is vit
al for slowing down the aging process.
* **Improved Physical Performance:**  Whether you're an athlete or just trying to keep up with daily activities, adequat
e sleep enhances reaction time, coordination, and overall physical performance.


**2. Mental & Cognitive Benefits:**

* **Improved Memory & Learning:** Sleep plays a critical role in consolidating memories - transferring information from
```

Now one has setup basic environment that can be used to learn how to consume LLMs using python and nodejs using a locally hosted ollama.

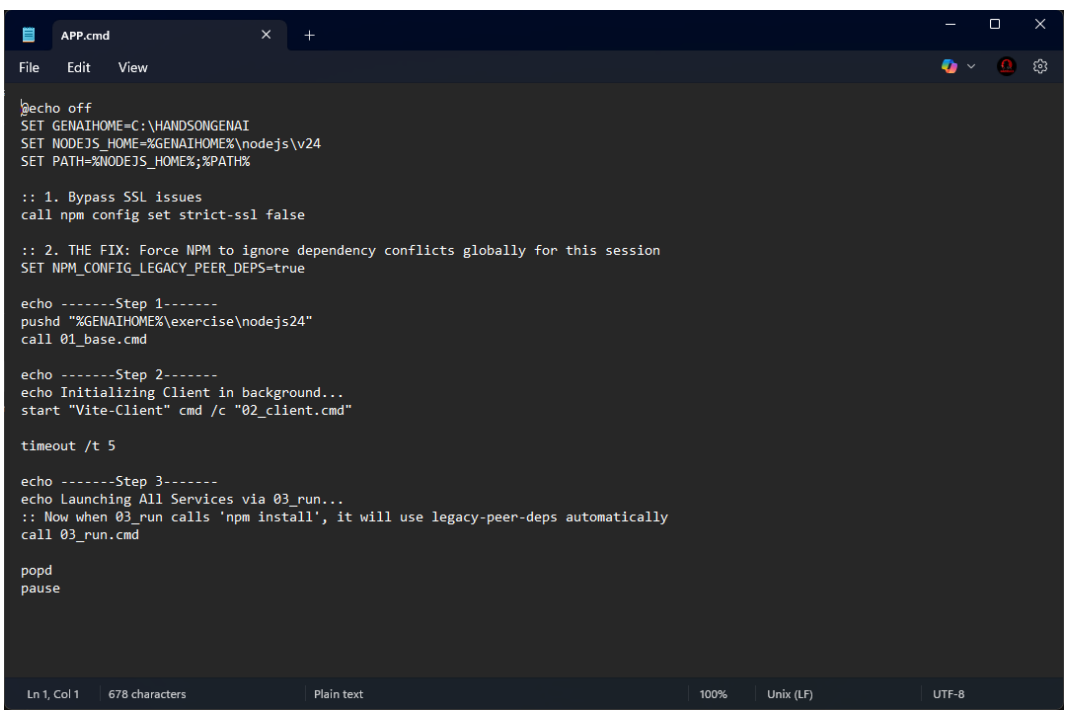
Next step would be to learn how to use a hyperscaler LLM like Gemini or OpenAI or Claude etc.

CREATE APPLICATION LAUNCHER SCRIPT

Create a *APP.cmd* in *C:\HANDSONGENAI* and paste the following lines of code in it:

```
@echo off
SET GENAIHOME=C:\HANDSONGENAI
SET NODEJS_HOME=%GENAIHOME%\nodejs\v24
SET PATH=%NODEJS_HOME%;%PATH%
```

```
:: 1. Bypass SSL issues
call npm config set strict-ssl false

:: 2. THE FIX: Force NPM to ignore dependency conflicts globally for this session
SET NPM_CONFIG_LEGACY_PEER_DEPS=true

echo -------Step 1-------
pushd "%GENAIHOME%\exercise\nodejs24"
call 01_base.cmd

echo -------Step 2-------
echo Initializing Client in background...
start "Vite-Client" cmd /c "02_client.cmd"

timeout /t 5

echo -------Step 3-------
echo Launching All Services via 03_run...
:: Now when 03_run calls 'npm install', it will use legacy-peer-deps automatically
call 03_run.cmd

popd
Pause
```



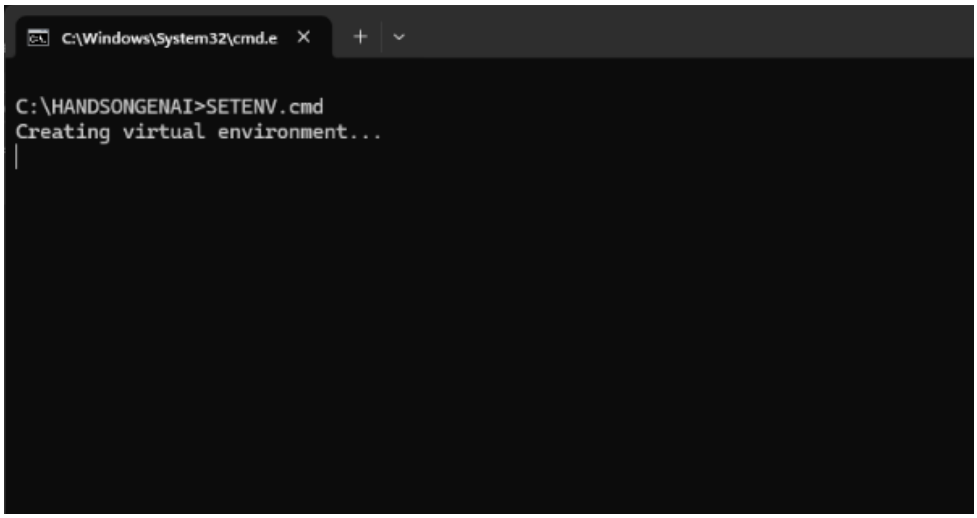
STEPS TO RUN THE SERVER/CLIENT

Follow the steps below to run the server/client. Make sure you have finished all the required setup mentioned above.

1. `Launch Ollama.exe and keep it running in the background.`
2. Double-click on `SETENV.cmd` (or) type the command as follows while on path *`C:\HANDSONGENAI`*

   ```
   SETENV.cmd
   ```

   This will launch the SETENV.cmd script we wrote.

3. The environment will set-up and calls APP.cmd after setting the environment. This will popup new CMD windows in parallel to run the server & client.
4. On first run, you are expected to confirm for installing vite packages on the CMD Window. Type 'Y' and hit enter.

```
C:\WINDOWS\system32\cmd.

(vpy313) C:\HANDSONGENAI\exercise\nodejs24\chatapp>cd client

(vpy313) C:\HANDSONGENAI\exercise\nodejs24\chatapp\client> npm create vite@latest . -- --template
react create-vite@8.2.0
Ok to proceed? (y) Y
```

```
C:\WINDOWS\system32\cmd.

(vpy313) C:\HANDSONGENAI\exercise\nodejs24\chatapp>cd client

(vpy313) C:\HANDSONGENAI\exercise\nodejs24\chatapp\client>npm create vite@latest . -- --template react

> client@0.0.0 npx
> create-vite . --template react

|
*  Current directory is not empty. Please choose how to proceed:
|  > Cancel operation
|    Remove existing files and continue
|    Ignore files and continue
-
```

5. Now click on `Cancel operation`. Use your *arrow keys* to navigate and *Enter* to choose an option.

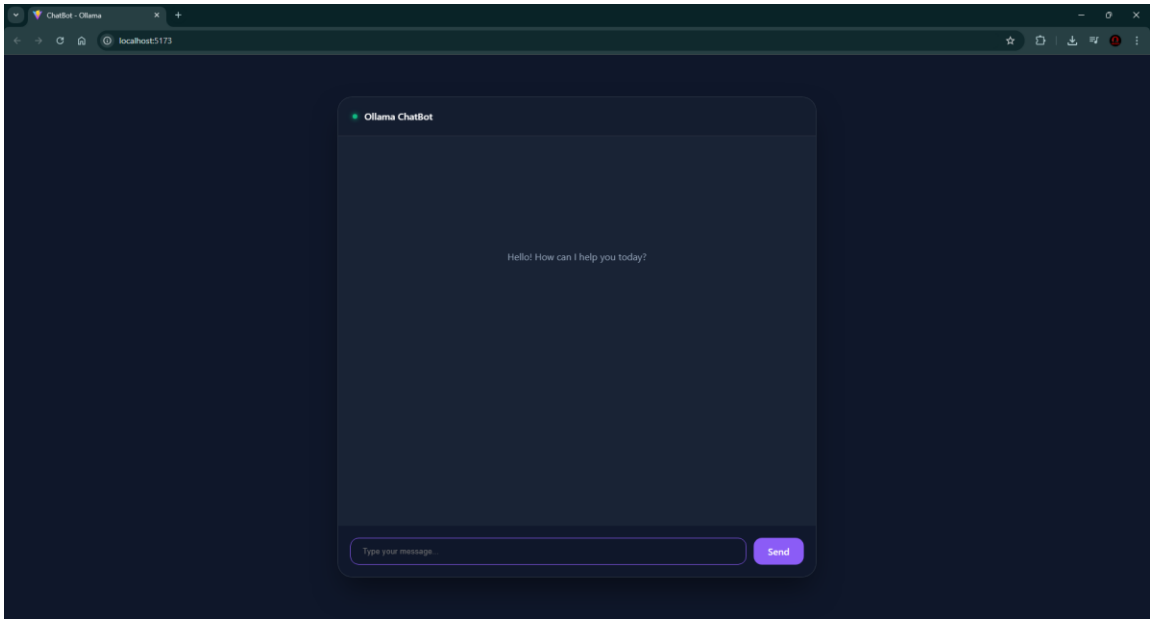
Server Window:

```
npm run dev

up to date, audited 174 packages in 1s

34 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

> agentic-backend@0.1.0 dev
> node src/index.js

Agent server running at http://localhost:3001
```

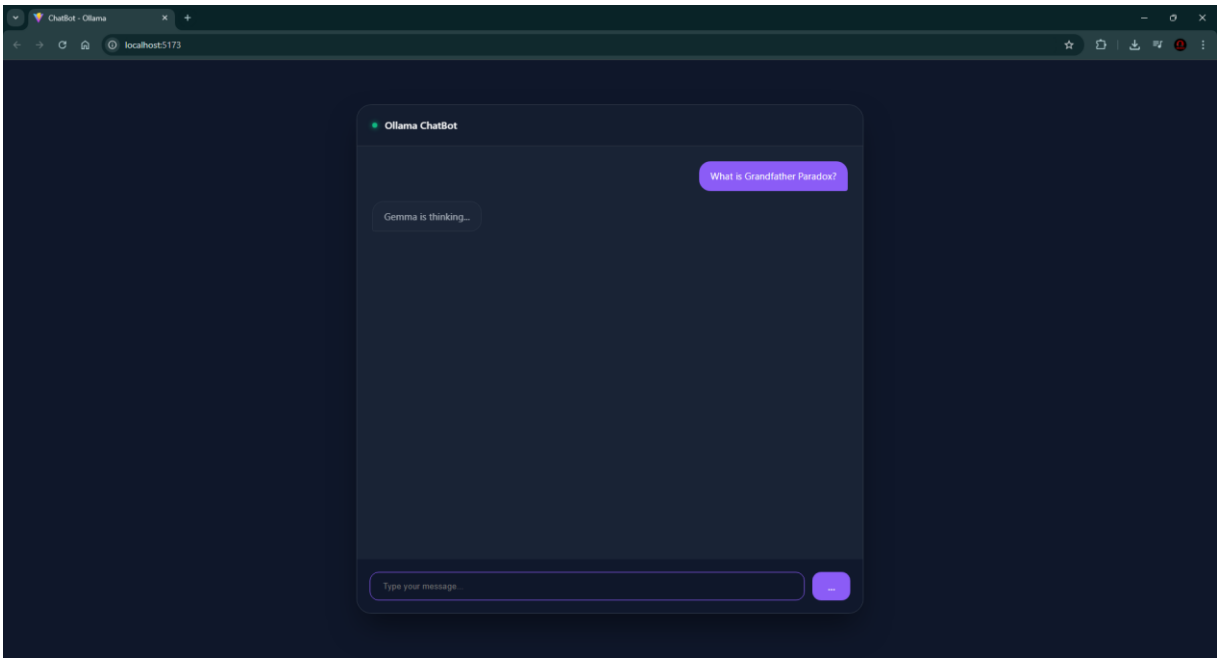
Client Window:

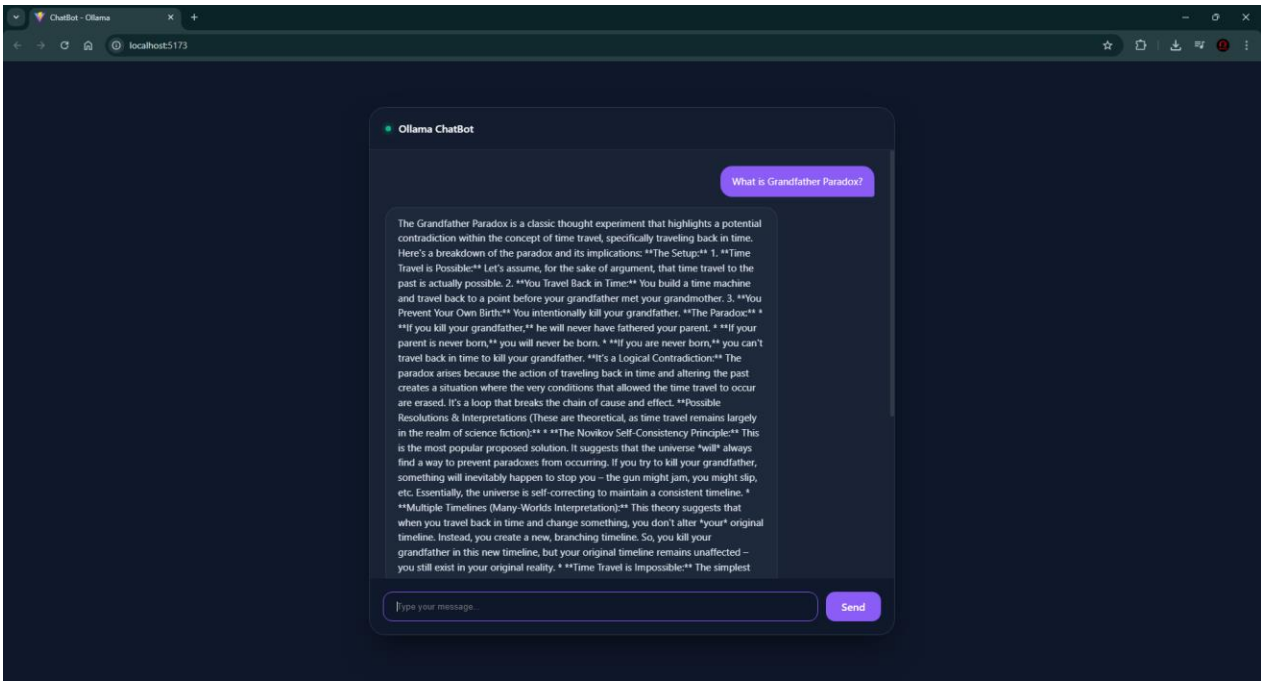
```
C:\WINDOWS\system32\cmd.

up to date, audited 158 packages in 1s

33 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

> client@0.0.0 dev
> vite


  VITE v7.3.1  ready in 1316 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
```

Now, the application is ready to use. Open a browser of choice and visit *http://localhost:5173/*



6. Enter your prompt and click on send. The response time depends on the network speed, and model selected.




ChatBot - Ollama
localhost:5173
Ollama ChatBot
What is Grandfather Paradox?
The Grandfather Paradox is a classic thought experiment that highlights a potential contradiction within the concept of time travel, specifically traveling back in time. Here's a breakdown of the paradox and its implications: **The Setup:** 1. **Time Travel is Possible:** Let's assume, for the sake of argument, that time travel to the past is actually possible. 2. **You Travel Back in Time:** You build a time machine and travel back to a point before your grandfather met your grandmother. 3. **You Prevent Your Own Birth:** You intentionally kill your grandfather. **The Paradox:** * **If you kill your grandfather,** he will never have fathered your parent. * **If your parent is never born,** you will never be born. * **If you are never born,** you can't travel back in time to kill your grandfather. **It's a Logical Contradiction:** The paradox arises because the action of traveling back in time and altering the past creates a situation where the very conditions that allowed the time travel to occur are erased. It's a loop that breaks the chain of cause and effect. **Possible Resolutions & Interpretations (These are theoretical, as time travel remains largely in the realm of science fiction):** * **The Novikov Self-Consistency Principle:** This is the most popular proposed solution. It suggests that the universe *will* always find a way to prevent paradoxes from occurring. If you try to kill your grandfather, something will inevitably happen to stop you – the gun might jam, you might slip, etc. Essentially, the universe is self-correcting to maintain a consistent timeline. * **Multiple Timelines (Many-Worlds Interpretation):** This theory suggests that when you travel back in time and change something, you don't alter *your* original timeline. Instead, you create a new, branching timeline. So, you kill your grandfather in this new timeline, but your original timeline remains unaffected – you still exist in your original reality. * **Time Travel is Impossible:** The simplest
Type your message...
Send